

Attention patterns are visible as matrices

Cross-attention, bidirectional self-attention, and causal self-attention use the same scoring idea with different masks.

Cross-attention

target queries source tokens

	s1	s2	s3	s4
t1	0.7	0.2	0.1	0.1
t2	0.1	0.2	0.3	0.9
t3	0.1	0.8	0.3	0.2

Rows are target positions; columns are source positions.



Self-attention

each token may read all tokens

	x1	x2	x3	x4
x1	0.9	0.4	0.2	0.5
x2	0.3	0.8	0.6	0.2
x3	0.5	0.4	0.9	0.3
x4	0.2	0.7	0.4	0.8

No mask: every query can score every key.



Causal self-attention

future positions are blocked

	x1	x2	x3	x4
x1	0.9	×	×	×
x2	0.6	0.8	×	×
x3	0.2	0.5	0.9	×
x4	0.3	0.2	0.6	0.8

Upper triangle is masked before softmax.

